Mohamed Sakr Structural Engineer · BIM & Digital Delivery

mo.sakr970@gmail.com +372 56711493 Tallinn /mosakr970 /mosakr970

Structural engineer with 10 years of experience on commercial, industrial, and residential projects across Estonia, Egypt, and Saudi Arabia including buildings up to 12,000 m^2 taken through Tekla Structures from concept model to issued shop drawings. Currently at ESPLAN in Tallinn, where alongside design work I have built custom parametric Tekla components and automation rules that cut repetitive modelling time by up to 30% plus Tekla macros tested on live project models. Holds an MSc from TalTech, with a graduation project completed at Aalto University. Bringing structural expertise and Tekla automation experience to BIM management and construction technology roles.

BIM AUTOMATION TOOLKIT

Single-file Tekla macros (C#/.NET, AI-assisted): Case studies results: github.io/tekla-automation-toolkit

HC Check: flags hollow-core panels and openings outside factory width bands before fabrication. 851 panels cross-checked against an independent implementation, zero disagreements.

Insert Tendons: models post-tensioning tendons and ducts from the designer's Excel into sloped, rotated and polybeam geometry. 59 beams, 424 parts, 11,282 points, zero failed; ~5 minutes vs the team estimation ~40hr.

Auto Dimension: grid-referenced, chained drawing dimensions that avoid existing annotations. 52 of 52 dimension points bound to the model.

EK Mark: gives identical assemblies the same mark. 1,356 of 1,357 pairs matched Tekla's own numbering.

Opening Marks: crosses out floor-plan openings. 37 openings on a 1,572-part plan, every line end within 1 mm.

CORE SKILLS

BIM & Modelling: Tekla Structures (custom components, automation rules, macros) | Revit Structure | AutoCAD | IFC & custom property sets | BIMcollab Zoom | Shop drawing | 3D coordination | Clash detection

Structural Tools: FEM design | Robot Structural | ETABS | SAP2000 | SAFE | STAAD.Pro | ABAQUS

Design Codes: Eurocode | ACI-318 | ASCE-7 | ECP-203 | ECP-201 | UBC/IBC

Project Skills: Structural coordination | Quality management | BOQ preparation | Technical reporting

Digital Development: Tekla macros (C#/.NET, AI-assisted) | IFC | open BIM formats | Scripting basics

PROFESSIONAL EXPERIENCE

Structural Engineer - BIM & Modelling, *ESPLAN* 07/2018 – Present | Tallinn, Estonia

Structural design and Tekla Structures modelling for residential towers, industrial facilities, commercial buildings, and public buildings, projects ranging from 2,000 to 12,000 m^2, from structural layout through coordinated, issued shop drawings.

- Delivered models and shop drawings in Tekla Structures for **Rocca towers** (~10,000 m^2), **Fortaco Factory** (~12,000 m^2), and **Lidl** stores.
- Coordinated structural models with architectural and MEP disciplines, resolving clashes at design stage.
- Performed quality review of structural deliverables against Eurocode requirements before issue.
- Built parametric custom components and automation rules in Tekla that cut model setup and drawing generation time, adopted as team standard workflow.
- Built five Tekla macros tested on live project models.
- Updated custom IFC property sets so element data matches downstream coordination and checking needs.

Structural Engineer, *Jawda Consultants* 02/2017 – 06/2018 | Cairo, Egypt

- RC analysis and design for four high-voltage substations (110/13.8 kV and 132/33 kV) for Saudi Electricity Co. National Grid KSA, in STAAD.Pro to ACI-318, ASCE-7 and UBC.
- 3D model coordination and clash detection with electrical, architectural and MEP disciplines.

Structural Engineer & Site Engineer, *Alhoda Consultant* 2014 – 2017 | Cairo, Egypt

- RC analysis and design for residential towers (G+11), wastewater treatment plants and steel structures using ETABS, SAP2000 and SAFE to ECP-203.

EDUCATION

Professional Development Programme, *kood/Jõhvi* 2024 – 2026 | Estonia

- Scripting, process automation, and digital workflow tooling.

Master of Structural Engineering, *TalTech University* 2020 – 2022 | Estonia

- Graduation project completed at Aalto University:
  "Numerical studies of composite beam behaviour under fire".

Bachelor of Civil Engineering, *Zagazig University* 2009 – 2014 | Egypt

CERTIFICATIONS & REGISTRATIONS

Disciplined Agile Scrum Master (DASM) · Project Management Institute (PMI) | 2023

Civil Engineer in Buildings & Structures, ESTQF Level 7 · Eesti Ehitusinseneride Liit | 2024

LANGUAGES

Arabic: Native • **English:** C1 Professional • **Estonian:** B1 Intermediate